

Urban Infrastructure Inspection System

Data Structures & Algorithms Final Presentation
Group 22 | CPT204 Spring 2026

| Project Overview

Primary Objectives

Develop a robust system for managing municipal assets by integrating high-efficiency sorting with precise pathfinding algorithms.

- ✓ Ranking candidates for urgent inspection.
- ✓ Optimizing routing across 1000 city nodes.
- ✓ Applying clean OO architecture.



| Data Infrastructure



Datasets A/B/C

1000 inspection candidates per dataset with diverse initial orderings (ordered, randomized, reverse).



Weighted Graph

City map consisting of 1000 nodes and 2600 undirected edges with distance weights.



CSV Integration

Decoupled CSV Readers parsing data into immutable Candidate and Edge records.



Task A: Sorting Performance

| Algorithm Implementation

- ⚡ **Bubble Sort:** Optimized with a 'swapped' flag for $O(n)$ best-case performance.
- 📦 **Merge Sort:** Guaranteed $O(n \log n)$ stability regardless of data input.
- 🚀 **Quick Sort:** Engineered with **Median-of-Three** pivot selection to prevent worst-case degradation.

Sorting visualization

Benchmark Analysis

Dataset B (Randomized) Avg. Execution Time (ms)

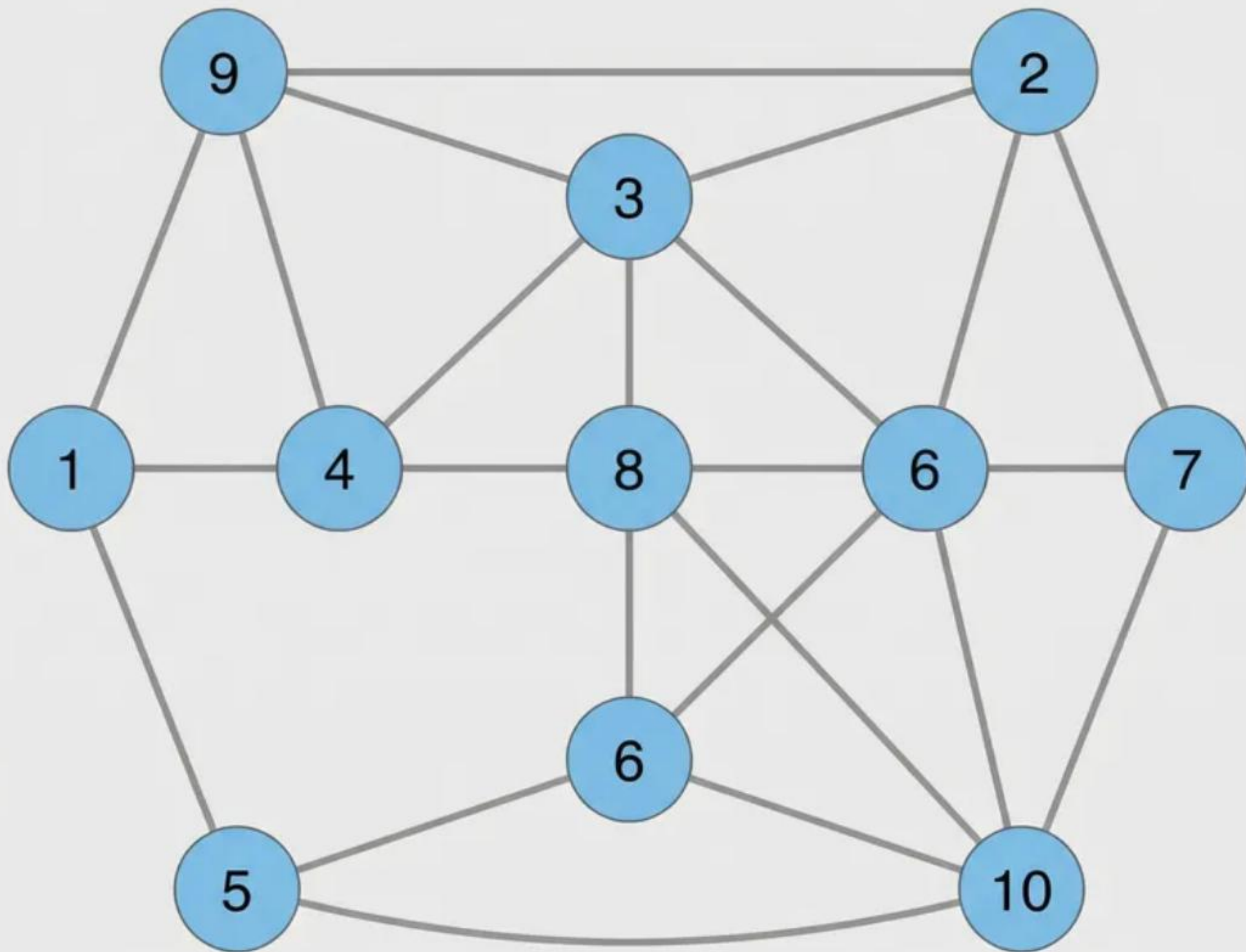


Results obtained using **JIT Warm-up mechanism to eliminate JVM cold-start jitter.*



Task B: Graph & Routing

| Dijkstra Pathfinding



- **</> Adjacency List:** Memory-efficient $O(V+E)$ representation for sparse city networks.
- **🧠 Priority Queue:** Min-Heap optimized Dijkstra at $O((V+E) \log V)$.
- **🔗 Via-Point Logic:** Robust path concatenation with duplicate node removal.

Shortest Path Case Results

Case	From / To / Waypoints	Path Highlights	Total Cost
1	A[1] to Self	Direct loop	0
2	A[1] to A[10]	L0001 → L0340 → ... → L0010	27
3	A[1] to B[1] via B[5]	L0106 → L0105 (Via) → L0106	39
4	A[1] to C[1] via B[5], C[5]	Passes B[5] then C[5] sequentially	48

| Object-Oriented Architecture



Encapsulation

Immutable Record-like classes for Candidates and Edges. Private final fields ensuring data integrity.



Polymorphism

Common Sorter interface allowing the system to swap Bubble, Quick, and Merge sorts dynamically.



Abstraction

Decoupled ShortestPathFinder interface, isolating routing logic from the core Graph topology.

Project Reflection

-  **Collaboration:** Jira/Trello task tracking with small Git commits for conflict-free merging.
-  **EDI Awareness:** Inclusive naming and planned Text-to-Speech support for visually impaired inspectors.
-  **Future Work:** Upgrading to A* Search and Time-Dependent Graphs for live traffic modeling.



Team Collaboration

Thank You

Questions & Answers | Group 22

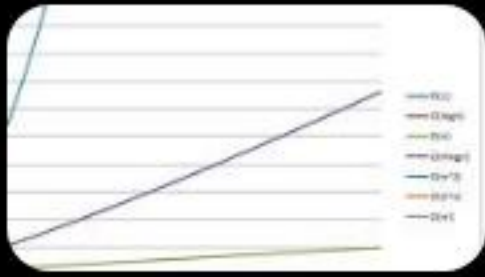
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| Image Sources



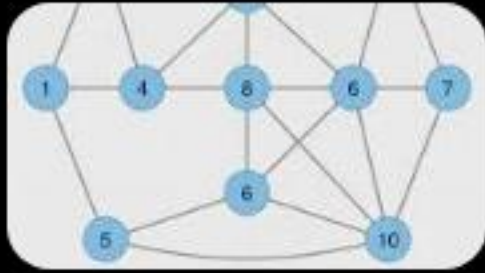
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Source: www.archivinci.com



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Source: visualgo.net



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Source: easy-peasy.ai



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